

Van Allen Magnetic & Plasma Surveyors (VAMPS): Attitude, Determination, & Control (ADC)

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SUBSYSTEM OVERVIEW

Subsystem Objectives

- Provide accurate attitude determination
- Maintain a stable and precise attitude
- Support GNC Operations

Subsystem Requirements

- **ADC-F-001:** Support pointing needs
- **ADC-F-006:** Fault tolerant safe mode
- **ADC-P-001:** Pointing knowledge of 4.29 arcseconds
- **ADC-P-004:** Reaction Wheel momentum capacity of 6Nm

Subsystem Drivers

- Achieve and maintain any given attitude (MISS-P-002: Downlinking data)
- Ensure sufficient solar charging to power spacecraft
- Radiation intense environment that surrounds the VAMPS mission

INTERFACES

- GNC: Shares thrusters, propellant tank, and OBC
- POW: Solar Panel pointing requirements for slew rate analysis
- CCD: Downlink pointing accuracy
- OPL: Payload instruments are in-situ sensors
- PLV: Separation tumbling rates inform de-tumbling design

VERIFICATION & VALIDATION

- Torque disturbance in accordance with individual computations and reaction wheel compatibility
- Propellant allocation independently verified across s/c
- The hardware components that we selected are compliant with our pointing requirements, and necessary radiation hardening to last for the duration of the mission
- All sensors meet the required knowledge error budget for the entirety of the VAMPS mission

FORMAL ANALYSIS

Torque Analysis: PDR used single literature scaled analysis but FDR replaces this with individually computed sources per Wertz SMAD Ch. 11.1:

Source	VAMPS 1 (Nm)	VAMPS 2 (Nm)	VAMPS 3 (Nm)
SRP	1.74×10^{-6}	1.74×10^{-6}	1.74×10^{-6}
Gravity Gradient	2.73×10^{-4}	9.16×10^{-5}	8.35×10^{-6}
Magnetic	5.28×10^{-6}	1.77×10^{-6}	1.61×10^{-7}
Total	2.80×10^{-4}	9.35×10^{-5}	1.03×10^{-5}

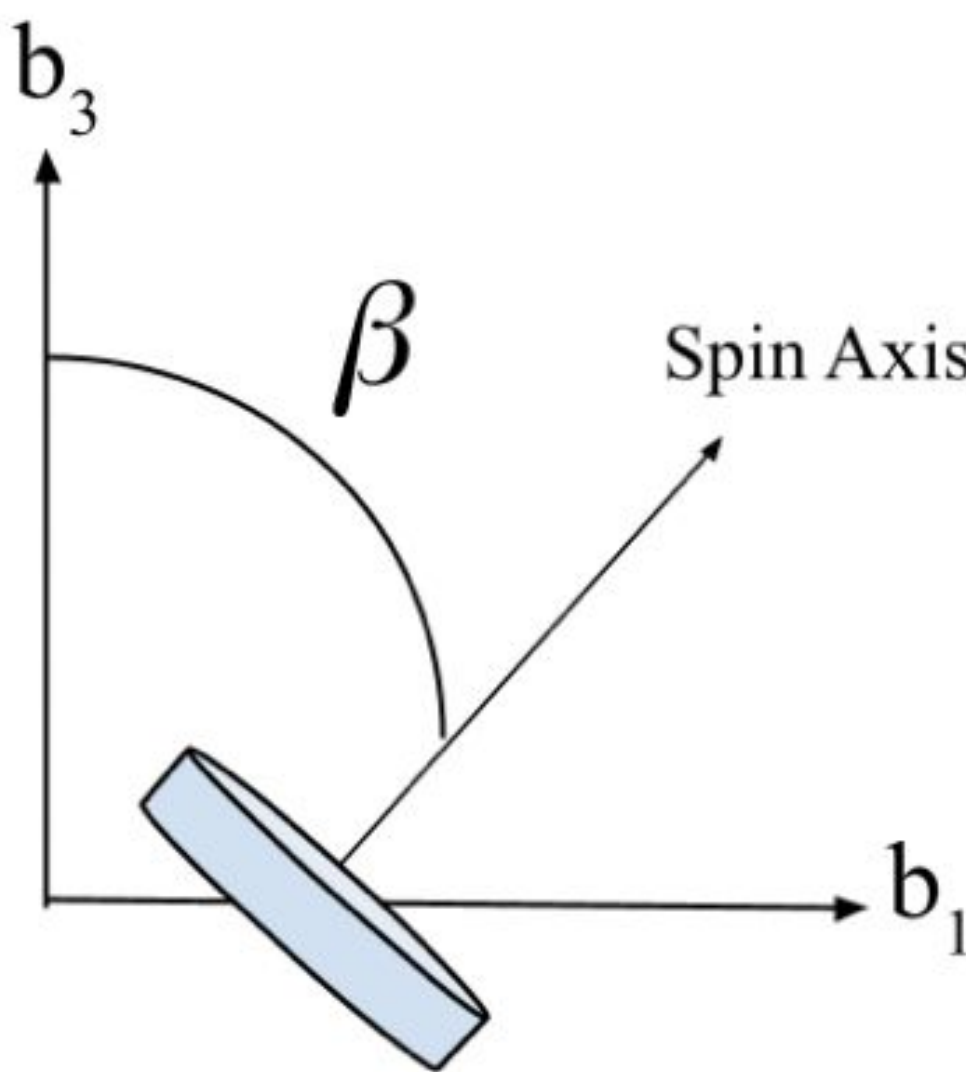
- Accumulated momentum over mission duration for propellant determination
- VAMPS 1, 2, and 3 require 7.11 kg hydrazine each to dump accumulated momentum

Cost, Mass, Power Budgeting

Component	Total Cost	Total Mass (Total)	Total Power (W) (Peak and SS)
L3Harris MR-401 LTR (ADC + GNC)	\$600,000	6.00 kg	10.0 0.0
ASTRO APS Star Tracker (ADC)	\$600,000	8.00kg	44.0 20.0
SSIRU IMU (ADC)	\$500,000	7.10 kg	43.0
Magnetometer (ADC)	\$6,000	3.0kg	1.0
Libra-6 (NRWA-T6) Reaction Wheels (ADC)	\$120,000	18.80 kg	117.2 15.3
Flight Computer (RAD750 3U Compact PCI) (ADC + GNC)	\$350,000	2.5 kg	5.0 10.0
Fox NavRIX GEO (GNC)	\$150,000	4 kg	10.0

Identifying Reaction Wheel Hardware

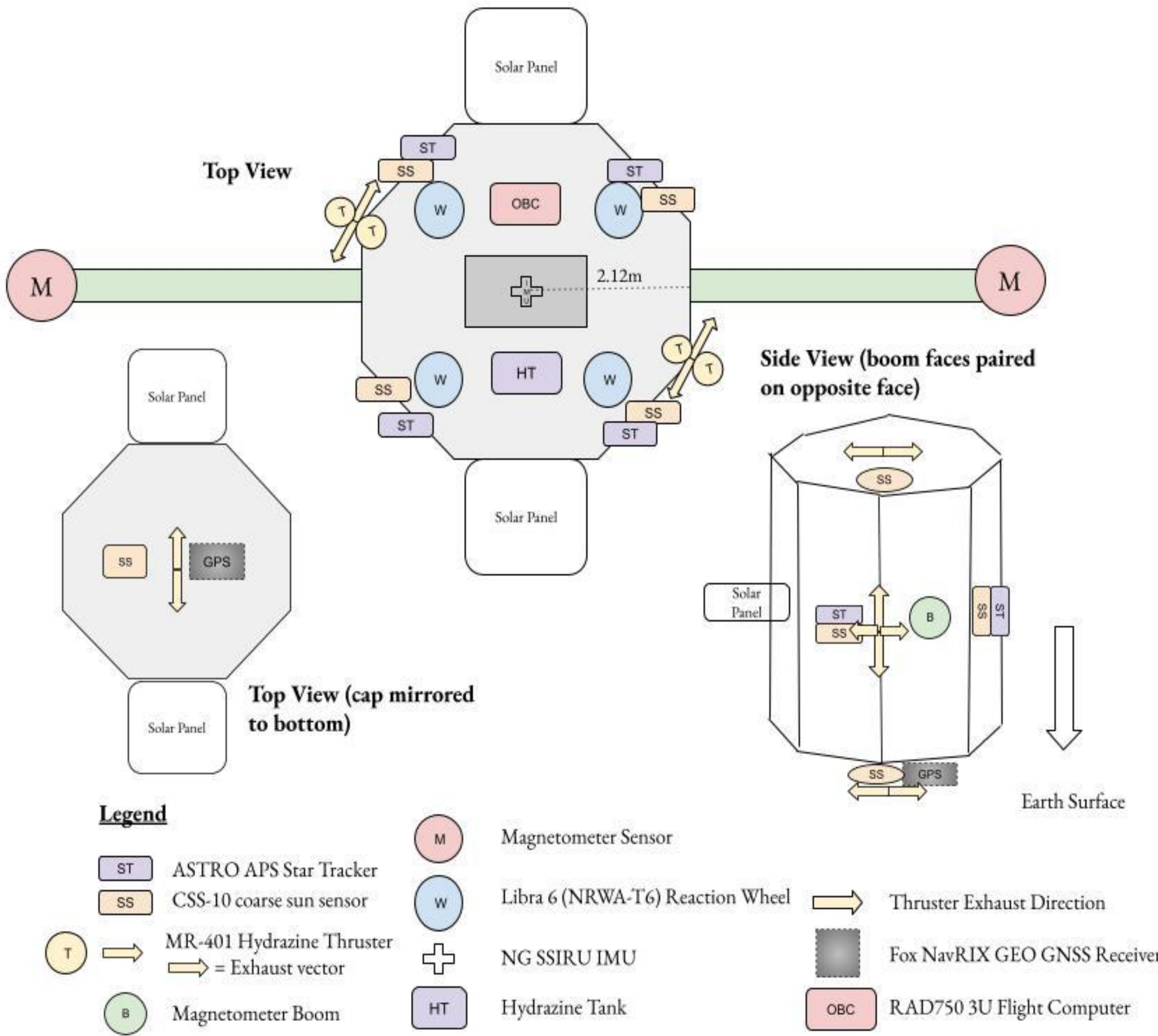
- Four reaction wheels are ideal for failure redundancy
- Reaction wheel selection is driven by pointing needs and slew rates
- Angular momentum vector is not aligned with rotation axis in pyramidal configuration, so placement and orientation drives hardware selection



$$H_{sat} = I_{sat}\omega_{sat}$$

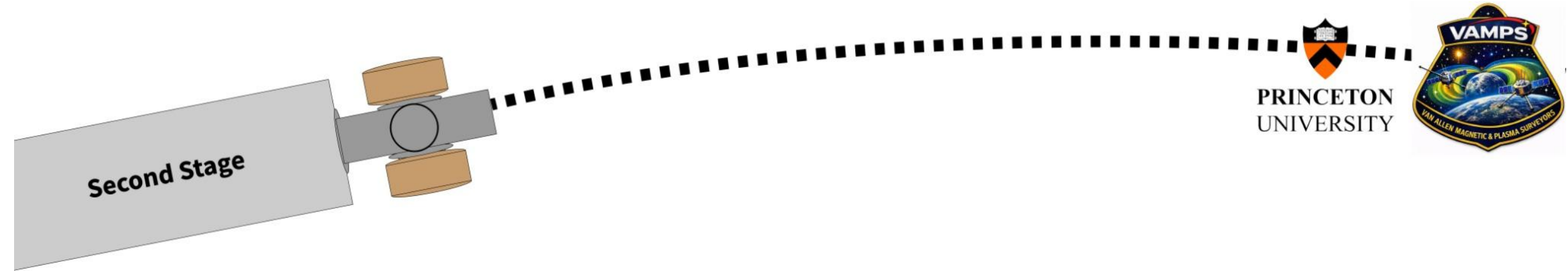
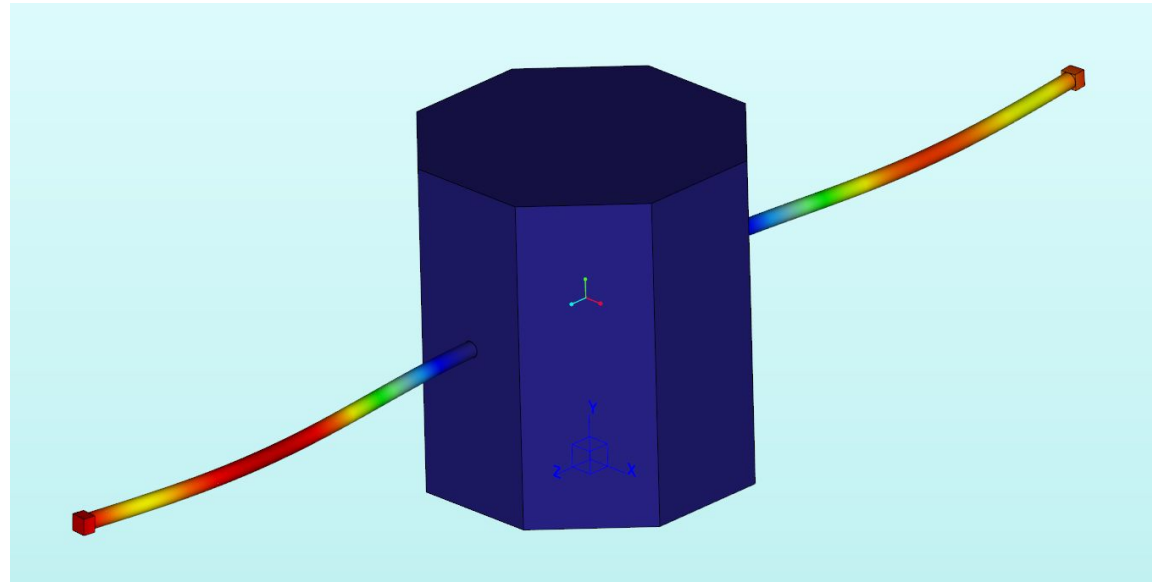
$$H_{RW} = Zh = \begin{bmatrix} \sin(\beta) & 0 & -\sin(\beta) & 0 \\ 0 & \sin(\beta) & 0 & -\sin(\beta) \\ \cos\beta & \cos\beta & \cos\beta & \cos\beta \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \end{bmatrix}$$

FINAL DESIGN CONFIGURATION



Resonant Frequency Analysis of Booms

- Natural frequency of magnetometer booms: 18.24 Hz (second order at 136.6 Hz)
- Natural frequency of EFW booms: 0.057 Hz
- Important to avoid the reaction wheels vibrating the spacecraft at these frequencies



VAMPS Deployment Sequence

